Meena Jagadeesan

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Summary

My research investigates machine learning ecosystems where an ML model (such as a language model) interacts with humans, companies, and other models. My goal is to steer how multi-agent interactions shape safety, performance, and society.

Academic Appointments

University of Pennsylvania *Philadelphia, PA*
ASSISTANT PROFESSOR, COMPUTER AND INFORMATION SCIENCE *Starting in July 2026*
- Faculty member of the ASSET Center for Trustworthy AI

Stanford University *Stanford, CA*
STANFORD AI LAB (SAIL) POSTDOCTORAL FELLOW *Sept 2025 - May 2026*
- Hosted by Tatsu Hashimoto and Sanmi Koyejo

Education

University of California, Berkeley *Berkeley, CA, USA*
PHD IN COMPUTER SCIENCE *August 2020 - May 2025*
- Advised by Michael I. Jordan and Jacob Steinhardt
- Affiliated with the Berkeley Artificial Intelligence Research Lab (BAIR)

Harvard University *Cambridge, MA, USA*
S.M. IN COMPUTER SCIENCE *August 2019 - May 2020*

Harvard University *Cambridge, MA, USA*
A.B. IN COMPUTER SCIENCE AND MATH, *summa cum laude* *August 2016 - May 2020*
- Secondary Field: Statistics
- Selected Honors: Phi Beta Kappa, Hoopes Prize, Detur Book Prize, Certificate of Distinction in Teaching

Selected Honors and Awards

Best Student Paper Award (with X. Hu, M. Jordan, and J. Steinhardt), Symposium on Foundations of Responsible Computing (FORC 2026)
Open Philanthropy AI Fellowship (2021-2025)
Paul and Daisy Soros Fellowship for New Americans (2020-2022)
CRA Outstanding Undergraduate Researcher Award (2020)
Siebel Scholarship (2019-2020)
Barry Goldwater Scholar (2018)
Intel Science Talent Search, 2nd Place in Basic Research (2016)

Publications

(* denotes equal contribution; α-β denotes alphabetical ordering)

JOURNAL ARTICLES

4. **Improved Bayes Risk Can Yield Reduced Social Welfare Under Competition**. *Quantitative Economics (Special Issue for ESIF: Economics and AI+ML meeting), to appear*.
Meena Jagadeesan, Michael I. Jordan, Jacob Steinhardt*, and Nika Haghtalab*.
3. **Safety vs. Performance: How Multi-Objective Learning Reduces Barriers to Market Entry**. *Proceedings of the National Academy of Sciences (PNAS), vol. 122, no. 42, e2510004122, 2025*.
Meena Jagadeesan, Michael I. Jordan, and Jacob Steinhardt.

2. **Accounting for AI and Users Shaping One Another: The Role of Mathematical Models (Survey Paper)**. *Transactions on Machine Learning Research (TMLR), 2025, p.2835-8856*.
(α-β) Sarah Dean, Evan Dong, Meena Jagadeesan, and Liu Leqi.
1. **Learning Equilibria in Matching Markets from Bandit Feedback**. *Journal of the ACM, 2023, Volume 70, Issue 3, Article no. 19, pp 1-46*.
Meena Jagadeesan*, Alexander Wei*, Yixin Wang, Michael I. Jordan, and Jacob Steinhardt.

Conference Proceedings

20. **Incentivizing High-Quality Content in Online Recommender Systems**. *Proceedings of the 6th Conference on Foundations of Responsible Computation (FORC), 2026*. Best Student Paper Award.
Xinyan Hu*, Meena Jagadeesan*, Michael I. Jordan, and Jacob Steinhardt.
19. **Impact of Decentralized Learning on Player Utilities in Stackelberg Games**. *Proceedings of the 41st International Conference on Machine Learning (ICML), 2024*.
(α-β) Kate Donahue, Nicole Immorlica, Meena Jagadeesan, Brendan Lucier, and Aleksandrs Slivkins.
18. **Feedback Loops With Language Models Drive In-Context Reward Hacking**. *Proceedings of the 41st International Conference on Machine Learning (ICML), 2024*.
Alexander Pan, Erik Jones, Meena Jagadeesan, Jacob Steinhardt.
17. **Clickbait vs. Quality: How Engagement-Based Optimization Shapes the Content Landscape in Online Platforms**. *Proceedings of the The Web Conference 2024 (WWW), 2024*.
(α-β) Nicole Immorlica, Meena Jagadeesan, and Brendan Lucier.
16. **Can Probabilistic Feedback Drive User Impacts in Online Platforms?**. *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS), 2024*.
(α-β) Jessica Dai, Bailey Flanigan, Nika Haghtalab, Meena Jagadeesan, and Chara Podimata.
15. **Supply-Side Equilibria in Recommender Systems**. *Proceedings of the 37th Conference on Neural Information Processing Systems (NeurIPS), 2023*.
Meena Jagadeesan, Nikhil Garg, and Jacob Steinhardt.
14. **Improved Bayes Risk Can Yield Reduced Social Welfare Under Competition**. *Proceedings of the 37th Conference on Neural Information Processing Systems (NeurIPS), 2023*.
Meena Jagadeesan, Michael I. Jordan, Jacob Steinhardt*, and Nika Haghtalab*.
13. **Competition, Alignment, and Equilibria in Digital Marketplaces**. *Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI), 2023*.
Meena Jagadeesan, Michael I. Jordan, and Nika Haghtalab.
12. **Performative Power**. *Proceedings of the 36th Conference on Neural Information Processing Systems (NeurIPS), 2022*.
(α-β) Moritz Hardt, Meena Jagadeesan, and Celestine Mendler-Dünner.
11. **Regret Minimization with Performative Feedback**. *Proceedings of the 39th International Conference on Machine Learning (ICML), 2022* .
Meena Jagadeesan, Tijana Zrnic, and Celestine Mendler-Dünner.
10. **Inductive Bias of Multi-Channel Linear Convolutional Networks with Bounded Weight Norm**. *Proceedings of the 35th Annual Conference on Learning Theory (COLT), 2022*.
Meena Jagadeesan, Ilya Razenshteyn, and Suriya Gunasekar.
9. **Individual Fairness in Advertising Auctions through Inverse Proportionality**. *Proceedings of the 13th Innovations in Theoretical Computer Science Conference (ITCS), 2022*.
(α-β) Shuchi Chawla and Meena Jagadeesan.
8. **Learning Equilibria in Matching Markets from Bandit Feedback**. *Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS), 2021*. NeurIPS 2021 Spotlight presentation (given to 10% of accepted papers).
Meena Jagadeesan*, Alexander Wei*, Yixin Wang, Michael I. Jordan, and Jacob Steinhardt.
7. **Alternative Microfoundations for Strategic Classification**. *Proceedings of the 38th International*

Conference on Machine Learning (ICML), 2021.
Meena Jagadeesan, Celestine Mendler-Dünner, and Moritz Hardt.

6. **Cosine: A Cloud-Cost Optimized Self-Designing Key-Value Storage Engine**. *Proceedings of Very Large Data Base Endowment (VLDB), 2021*.
Subarna Chatterjee, Meena Jagadeesan, Wilson Qin, and Stratos Idreos.
5. **Multi-Category Fairness in Sponsored Search Auctions**. *Proceedings of the 3rd ACM Conference on Fairness, Accountability and Transparency (FAT*), 2020*.
Christina Ilvento*, Meena Jagadeesan*, and Shuchi Chawla.
4. **Individual Fairness in Pipelines**. *Proceedings of the 1st Conference on Foundations of Responsible Computation (FORC), 2020*.
(α-β) Cynthia Dwork, Christina Ilvento, and Meena Jagadeesan.
3. **Understanding Sparse JL for Feature Hashing**. *Proceedings of the 33rd Annual Conference on Neural Information Processing Systems (NeurIPS), 2019*. NeurIPS 2019 Oral presentation (given to 3% of accepted papers).
Meena Jagadeesan.
2. **Simple Analysis of Sparse, Sign-Consistent JL**. *Proceedings of the 23rd International Conference on Randomization and Computation (RANDOM), 2019*.
Meena Jagadeesan.
1. **Varying the Number of Signals in Matching Markets**. *Proceedings of the 14th International Conference on Web and Internet Economics (WINE), 2018*.
Meena Jagadeesan* and Alexander Wei*.

Other Publications

3. **From Worst-Case to Average-Case Analysis: Accurate Latency Predictions for Key-Value Storage Engines**. *Proceedings of the ACM International Conference on Management of Data (SIGMOD), 2020 (2-Page Abstract)*. 1st Place at SIGMOD Student Research Competition.
Meena Jagadeesan* and Garrett Tanzer*.
2. **Dyson's Partition Ranks and their Multiplicative Extensions**. *The Ramanujan Journal, Vol. 45, Issue 3, pp. 817–839, 2018*.
(α-β) Elaine Hou and Meena Jagadeesan.
1. **Mobius Polynomials of Face Posets of Convex Polytopes**. *Communications in Algebra, Vol. 44, Issue 11, pp. 4945-4972, 2016*.
Meena Jagadeesan and Susan Durst.

Manuscripts under Submission

7. **Power and Limitations of Aggregation in Compound AI Systems**. *arXiv preprint*.
Nivasini Ananthakrishnan and Meena Jagadeesan.
6. **Deployment-Centered Evaluation: Predicting Query-Level Rejection Risk in a Clinical LLM System**. *Under submission*.
Alyssa Unell, Miguel Angel Fuentes Hernandez, Brenna Li, Bridget Lin, Meena Jagadeesan, Sanmi Koyejo, Nigam Shah.
5. **Metric Match: An Ensemble-Based Approach to Evaluating LLM Judge Reliability**. *Under submission*.
Alyssa Unell, Natalie Dullerud, Naomi Boneh, Meena Jagadeesan, Tatsunori Hashimoto, Nigam Shah, Sanmi Koyejo.
4. **Strategic Feature Selection and Regularization**. *Under submission*.
Jivat Neet Kaur, Pratik Patil, Divya M Shanmugam, Emma Pierson, Michael I. Jordan, Nika Haghtalab, Meena Jagadeesan*, Ahmed Alaa*, Serena Lutong Wang*.
3. **LLM Detection as an Intervention: Downstream Impact under Strategic User Behavior**. *Under submission*.

Meena Jagadeesan, Tatsunori Hashimoto, Jon Kleinberg.

2. **Breaking Algorithmic Collusion in Human-AI Ecosystems**. *arXiv preprint*.
   Natalie Collina, Eshwar Ram Arunachaleswaran, Meena Jagadeesan.
1. **Flattening Supply Chains: When do Technology Improvements lead to Disintermediation?**. *arXiv preprint*.
   (α-β) S. Nageeb Ali, Nicole Immorlica, Meena Jagadeesan, Brendan Lucier.

Theses

2. **Steering Machine Learning Ecosystems of Interacting Agents**. *PhD Thesis (CS Department at UC Berkeley)*.
   Advised by Michael I. Jordan and Jacob Steinhardt.
1. **The Performance of Johnson-Lindenstrauss Transforms: Beyond the Classical Setting**. *Undergraduate Thesis (Harvard College)*. Awarded Hoopes Prize.
   Advised by Prof. Jelani Nelson.

Talks

Seminars and Invited Talks

2026: *Stanford RAIN Seminar* (4/7/26); *Simons Institute Workshop on Agency in Collaborative Learning* (4/13/26)

2025: *University of Oxford, Nuffield Econometrics Seminar* (6/20/25); *Rutgers/DIMACS Theory of Computing Seminar* (10/15/25); *Rutgers Business School MSIS Seminar* (10/16/25); *INFORMS Annual Meeting, Modern AI and Marketplaces Session* (10/26/25)

2024: *Cornell Theory Seminar* (1/22/24); *UC Berkeley Economics Department, Theory Lunch* (3/6/24); *INFORMS Optimization Society Conference* (3/23/24); *Max Planck Institute Kaiserslautern, Next 10 in AI Series* (9/18/24); *Cornell ORIE Young Researchers Workshop* (10/9/24); *INFORMS Annual Meeting, How Competition Can Disrupt Machine Learning Session* (10/20/24); *INFORMS Annual Meeting, Learning in Online Platforms Session* (10/21/24)

2023: *Stanford University Rising Stars Workshop in Management Science and Engineering* (5/2/23); *MIT Reading Group on Human and Machine Decisions* (6/26/23); *Brookings Center on Regulation and Markets Seminar on AI, Economics, and Public Policy* (6/29/23); *Microsoft Research New England ML Ideas Seminar* (8/14/23); *INFORMS Annual Meeting, Learning and Mechanism Design Session* (10/17/23)

2022: *Workshop on Algorithms for Learning and Economics (WALE)* (6/15/22); *Workshop on Algorithms for Learning and Economics (WALE)* (6/16/22); *INFORMS Revenue Management and Pricing (RMP) Workshop* (6/22/22); *Northwestern CS Seminar & Institute for Data, Econometrics, Algorithms, and Learning (IDEAL) Seminar* (9/6/22, 9/7/22); *INFORMS Annual Meeting, Responsible, Ethical, and Socially Aware Operations Session* (10/16/22);

Pre 2022: *MIT Algorithms & Complexity Seminar* (4/7/21); *Google Research Algorithms Seminar* (5/20/21); *Microsoft Research MLO Group Seminar* (6/24/20); *Algorithmic Game Theory Mentoring Workshop at ACM EC* (6/15/20); *INFORMS Annual Meeting, Market Algorithms Session* (11/11/20); *University of Wisconsin-Madison Theory Seminar* (5/17/19);

Other

Contributed Conference and Workshop Talks: *FORC* (6/14/24); *ESIF-AIML* (8/13/24, 8/14/24); *ITCS* (2/1/22); *COLT* (7/3/22); *ICML* (7/21/22); *FORC* (6/10/21); *ICML* (7/21/21); *ACM FAT** (1/29/20); *RANDOM*

(9/21/19); *NeurIPS* (12/12/19); *Workshop on Frontiers of Market Design at ACM EC* (6/22/18); *WINE* (12/17/18)

Industry Experience

Microsoft Research — *Cambridge, MA*
RESEARCH INTERN — *Summers 2023 and 2024*
- Mentors: Nicole Immorlica and Brendan Lucier (Economics and Computation Group in MSR New England)

Microsoft Research — *Redmond, WA*
RESEARCH INTERN — *Summer 2020*
- Mentors: Suriya Gunasekar and Ilya Razenshteyn (Machine Learning and Optimization Group in MSR AI)

Microsoft — *San Francisco, CA*
SOFTWARE ENGINEER/PROGRAM MANAGER INTERN — *Summer 2018*

Teaching

Instructor for CIS 7000 at UPenn — *Fall 2026 (upcoming)*
- Will teach graduate-level course titled "Machine Learning Ecosystem".

Guest Lecture in Cornell Networks and Markets Course — *April 10th, 2024*
- 75 minute guest lecture on my research on how machine learning disrupts online marketplaces. I gave this guest lecture in CS 5854/ORIE 5138: Networks and Markets at Cornell Tech, taught by Professor Nikhil Garg.

Graduate Student Instructor for UC Berkeley Stat 157 — *Jan. 2023 - May 2023*
- Teaching assistant for undergraduate-level course on Forecasting in the Statistics department at UC Berkeley, taught by Prof. Jacob Steinhardt.

Graduate Student Instructor for UC Berkeley CS 281A — *Aug. 2021 - Dec. 2021*
- Teaching assistant for graduate-level introductory course on Statistical Learning Theory in the Computer Science department at UC Berkeley, taught by Prof. Moritz Hardt and Prof. Benjamin Recht.

Teaching Fellow for Harvard CS 61 — *Sept. 2018 - Dec. 2018*
- Teaching assistant for Harvard's introductory systems programming class for computer science undergraduates, taught by Prof. Eddie Kohler. Awarded a Certification of Distinction in Teaching.

Service

EXTERNAL SERVICE

Program Committee Member/Reviewer/Sub-Reviewer — *2019-*
- *Area Chair:* NeurIPS 2026, ICML 2026
- *Conference reviewing:* EAAMO 2026, SODA 2026, ITCS 2026, NeurIPS 2025, ICML 2024, FORC 2024, ACM FAccT 2024, The Web Conference (WWW) 2024, SODA 2024, NeurIPS 2023, ICALP 2023, ICML 2023, ICLR 2023, AISTATS 2023, ICML 2022, ACM FAccT 2022, ICLR 2022, NeurIPS 2021, ICML 2021, ACM FAccT 2021, STACS 2021, ITCS 2021, SOSA 2021
- *Workshop reviewing:* AAAI 2024 EcoSys Workshop, MATCHUP 2022, NeurIPS 2021 Workshop on Strategic ML
- *Journal reviewing:* ACM Transactions on Economics and Computation, Proceedings of the National Academy of Science, Management Science, Journal of AI Research

Co-organizer of Trustworthy AI Day at WALE 2026 — *July 2026*
- Co-organizing a day of invited talks and panel discussion focused on Trustworthy AI at the Workshop on Algorithms for Learning and Economics (WALE)

Workflow Chair for ACM EC 2025 — *July 2025*
- Member of the organizing committee for EC 2025. I organized virtual streaming of contributed talks at the main conference to enable broader access and participation.

Co-organizer of a Session at the INFORMS 2024 Meeting — *Oct 2024*
- I co-organized a session on "How Competition Can Disrupt Machine Learning" in the Revenue Management and Pricing Section at the INFORMS Annual Meeting. The session showcased 5 recent papers on this topic.

Mentor at Learning Theory Alliance Mentorship Workshop — *Nov 2023, Nov 2025*
- Served as a volunteer mentor at a virtual mentorship workshop focused on learning theory.

Panelist in a NeurIPS 2022 Tutorial — *Dec 2022*

- The NeurIPS 2022 tutorial, organized by Chara Podimata, focused on incentive-aware machine learning. I was one of five panelists in a 30 minute panel discussion on incentive-aware ML and the direction of the field.

Internal Service

Postdoc reviewer for SAIL Postdoctoral fellowship *2026*

- Reviewed and evaluated applications for the SAIL Postdoctoral Fellowship program.

Co-President of Women in CS and EE (WICSE) at UC Berkeley *2022-2023*

- WICSE is a community of UC Berkeley graduate students focusing on creating a welcoming and supportive environment for graduate studies. I was one of the two co-presidents of this organization.
- *Responsibilities:* As a co-president, I led and worked with a board of 10 graduate student volunteers. We organized weekly community lunches, social events, a day-long outreach event for a local Girl Scouts troops, and an in-person conference for Stanford and Berkeley students.

Co-organized the Berkeley-Stanford Women in EECS Meetup *April 29th, 2023*

- The 9th annual research meetup for Stanford and Berkeley EECS PhD students took place at UC Berkeley. The meetup consisted of a faculty keynote speaker, panels with faculty and industry researchers, student spotlight talks, and a celebration of the 45th anniversary of UC Berkeley WICSE.

Co-organizer of Rec Sys + Society Meeting at UC Berkeley *2022-2023*

- The weekly lunch meeting brought together researchers across the Berkeley AI Research Lab who study the societal implications of recommender systems. Each week, a different speaker presented on a topic in this domain and led a discussion. I was one of two co-organizers for these lunch meetings.

Mentor for Undergrad Mentorship Program at UC Berkeley *2022-2023*

- Mentored two promising undergraduates from underrepresented groups to help them get started in pursuing a career in AI. Mentors provide general career and academic advice.

Student Member of UC Berkeley PhD Admissions Committee *2021-2022*

- Reviewed and evaluated applications to the UC Berkeley EECS PhD program directed to the AI Theory subfield.